Fiscal Year:	FY 2009	**Task Last Updated:**	FY 02/27/2009
PI Name:	Mitchell, Cary Ph.D.		
Project Title:	(NSCORT) for Advanced Life Support at Purdue/Howard/AAMU - Integrated Systems Group		
Division Name:	Human Research		
Program/Discipline:	ADVANCED HUMAN SUPPORT TECHNOLOGIES		
Program/Discipline--Element/Subdiscipline:	ADVANCED HUMAN SUPPORT TECHNOLOGIES--Advanced life support		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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City:	West Lafayette	**State:**	IN
Zip Code:	47907-2010	**Congressional District:**	4
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	NSCORT
Start Date:	12/01/2002	**End Date:**	11/30/2009
No. of Post Docs:	0	**No. of PhD Degrees:**	1
No. of PhD Candidates:	1	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JSC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:	NOTE: Received NCE to 11/30/2009 per PI (1/09) NOTE: Received NCE to 11/30/2008 per PI (11/07)		
Key Personnel Changes/Previous PI:	0		
COI Name (Institution):	Gardner, Guy (No Longer at Purdue) Pekny, Joe (Purdue University) Chiu, George (Purdue University) Yao, Bin (Purdue University) Yih, Yuehwern (Purdue University) Orcun, Seza (Purdue University)		
Grant/Contract No.:			
Performance Goal No.:			
Performance Goal Text:			

Task Description:	Throughout the life of ALS NSCORT, the systems group researchers matured from learning and exploratory mode, which can be mainly characterized by proof-of-concept studies to build a viable toolbox of systems analysis methodologies suitable for ALS analysis, e.g. SimOpt (SIMulation based OPTimization), Markov Theory and Model Predictive Control, towards well structured and coordinated efforts to improve their readiness level for repeated use in answering questions that matters to NASA and ALS community at large. As systems group was undergoing through its voyage within ALS NSCORT, the systems group's analysis and modeling scope has been expanded to all ALS technologies and to adopt lifecycle management view to study 15-20 year Mars Base ALS system evolution which recognizes the fact that Mars Base ALS system will not stay as it was deployed at the very first mission. Unfortunately, our plans and resources had to be revised per early termination notice as NASA was undergoing through major re-organization and re-prioritization to maximize the return on NASA's investment in ALS NSCORT by focusing on completing the matured on going tasks and retiring the new initiatives such as, the proof-of-concept study that has been initiated in collaboration with Envision Center at Purdue University on visualizing the evolution of Mars Base and its ALS system. Throughout 4 years of ALS NSCORT program, systems group has trained 6 graduate students, engaged 11 researchers (3 Post-Docs, 4 original PIs, 4 additional researchers) with ALS NSCORT sponsored systems related projects, contributed to SIMA Lunar Challenge and BVAD, reached out to over 400 undergraduate students by challenging them with ALS driven projects in ChE 450 (Chemical Engineering Design Course) and voluntary undergraduate research projects (Rebecca Alway-Cooper of Chemical Engineering), supported other projects and programs within ALS NSCORT by giving invited lectures at the McKenzie Career Center to Mr. Martin's high school class, editing a "baby BVAD" intended to be used in the project "Equivalent Systems Mass Analysis of Plant Growth" with high school students lead by outreach group of ALS-NSCORT, leading and supporting center-wide project management, roadmap development, process map revisions and ESM evaluation of ALS-NSCORT technologies initiatives and activities. ALS NSCORT researchers on the average gave 3 TIM/SIMA telecons per year, generated 18 conference papers (9 ICES, 5 Habitation, 2 AICHE, 3 other), 1 peer reviewed journal paper. We expect that ALS NSCORT scholarly products will continue to emerge in the next couple of years at various journals (there are 6 papers in progress in our pipeline). We, in collaboration with ALS NSCORT management, have further initiated activities: 1) to archive knowledge generated throughout our ALS NSCORT program within the Purdue University Libraries, 2) to reflect our experience through a lessons learned white paper which will be ready for dissemination in couple of months after the early termination of ALS NSCORT. We believe that these initiatives will enable us to respond to large scale ALS research program opportunities rapidly as they emerge. Seza Orcun Integrated Systems Group – Focus Area Lead
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Development of computer models for control of complex processes
Task Progress:	Simulation of Control Policies for the Health-Monitoring System of a Closed System for Human Habitation of Space A simulation is developed to assist decision-making in choosing control policies in a closed environment. This closed environment consists of a simplified representation of the water subsystem based on the documentation by the National Aeronautics and Space Administration (NASA). The water requirement (consumption) and generation are based on the crewmembers' demographics, activity schedules as well as intensity of each activity. The health of this system is the result of the intricate interaction between the crewmembers as well as the water subsystem. The primary measure of the health of the system is the "state" representation of the system. This representation is a conglomeration of various hourly assessments of different aspects of the system. These aspects include the deterministic as well as the stochastic nature of the system. Due to the potentially vast number of possible states in the entire state space, state reduction techniques as well as assessments for infeasible states are applied for state reduction. A reward system is designed for allocating reward/penalty values to system states that satisfy predetermined conditions. Due to the strong stochastic characteristics and the "memoryless" property of the system, the transition of system state can be modeled as a Markov process. Transition probabilities are obtained through the simulation of the Markovian model. Capturing "true" state transition probabilities has been a challenge to researchers. In this research, a methodology is under development to minimize the number of data points (or simulation hours) while obtaining the transition probabilities that are closest to the system's steady state probabilities by observing the percentage increase of states visits as well as probabilities convergence based on discrete consecutive time periods. These probabilities, together with the reward values are formulated into appropriate mathematical representation and are solved by applying the Policy Iteration (PI) algorithm. The outcome from PI is an indication of the "best" policy to be applied when the system is in a given state. Various scenarios with different system design, magnitude of stochastic properties and system modules such as the addition of a crop subsystem can be studied to evaluate the benefits of applying this methodology. A comparison of a deterministic system versus a stochastic system is also done. Results show that the stochastic system out-performs the deterministic system in terms of states captured, states missed, overall rewards, as well as deficiency values. The simulation was done in three stages: Data Collection/State space exploration, System Evaluation with Dynamic Policy and System Evaluation with Fixed Policy. The Data Collection stage was performed through 24000 ticks (simulation hours) of simulation time. With the use of the Random Walk (Drunkard's Walk) technique, 120 states were captured. Outcomes from Stage 2 are compared against a system that is not modeled as a Markov process and uses a baseline policy regardless of system states. Due to the use of a controlled input of common random number stream, a paired t-test is used to compare the outcomes of the simulation with best policies and one with a fixed policy. Statistical analysis shows that the system with best policies performed at a level that is 34.7% more superior (in terms of reward value) than the system with a fixed policy. Based on the proposed methodology, the next scenario to be tested consists of a study of dynamically changing the control policies based on changes in the system and its environment. A set of "best policies" is obtained and applied based on limited knowledge of the system. As the knowledge of the system increases, feedback is used and the proposed

	algorithm will generate a new set of “best policies” based on the new knowledge of the system. Results are pending for this study.
Bibliography Type:	Description: (Last Updated: 04/16/2020)
Articles in Peer-reviewed Journals	Chiam TC, Yih Y, Mitchell CA. "Control policies for a water-treatment system using the Markov Decision Process. Part 2: Simulation and analysis." Habitation - An International Journal for Human Support Research. (Accepted) November 2008. , Nov-2008
Articles in Peer-reviewed Journals	Chiam TC, Yih Y, Mitchell CA. "Control policies for a water-treatment system using the Markov Decision Process. Part 1: Mathematical formulation." Habitation - An International Journal for Human Support Research. (Accepted), November 2008. , Nov-2008
Articles in Peer-reviewed Journals	Ang CH, Yih Y. "A simulated approach to minimize water supply, water storage capacity and water treatment capacity requirements in an Advanced Life Support system for Mars missions." Habitation. 2008;11(4):173-83. , Sep-2008